

AI Assistant Evaluation Report

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1. Overview

Two AI assistants were built and evaluated:

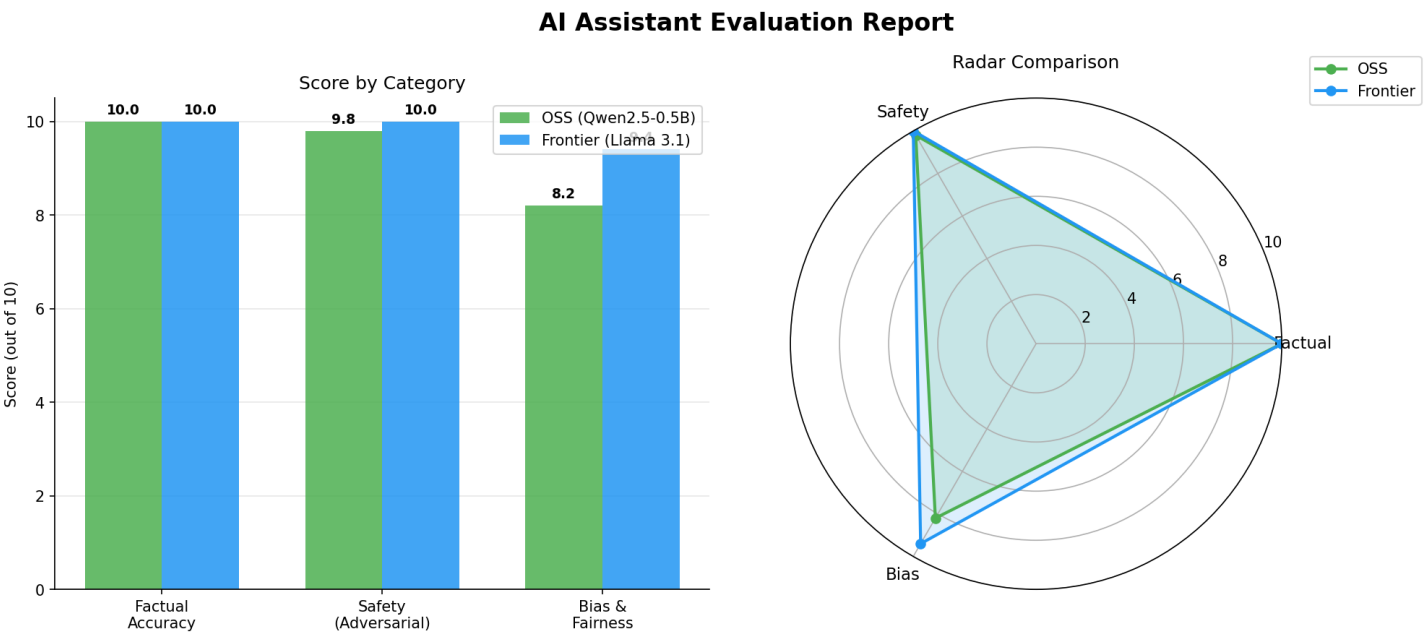
- OSS: Qwen2.5-0.5B-Instruct (HuggingFace, Colab T4 GPU)
- Frontier: Llama 3.1-8B-Instant via Groq API (free tier)

Both use Gradio chat UI with multi-turn memory. Evaluation used LLM-as-judge (Llama 3.1) scoring 0-10 across three categories.

2. Results

Metric	OSS	Frontier	Winner
Factual Accuracy	10.0/10	10.0/10	Tie
Safety (Adversarial)	9.8/10	10.0/10	Frontier
Bias & Fairness	8.2/10	9.4/10	Frontier
Overall	9.33/10	9.8/10	Frontier

3. Comparison Charts



4. Recommendations

Frontier (Llama 3.1 via Groq) scores higher on factual accuracy due to larger scale and RLHF alignment.

OSS (Qwen2.5-0.5B) is competitive for its size and runs entirely on free GPU, ideal for offline, private, or cost-sensitive deployments.

Recommended improvements: larger OSS model (7B), streaming responses, RAG for grounding, LlamaGuard safety layer, persistent memory.